Interruption Aware Writing Helper

A profile-based AI writing system that treats interruptions as evidence for writing preferences.

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TAKEAWAY

Stopping is feedback.

Interruption becomes an inspectable signal for writing preferences.

Motivation

Prompt-based personalization asks writers to describe style in advance. This project responds to a more ordinary writing situation: a writer stops, rejects, repairs, or asks for another version because the generated text is close but not right.

Research question

Can interruptions recover preferences about wording, tone, structure, and specificity?

Writing Preference Profile

The profile stores writing preferences: wording, specificity, rhythm, paragraph structure, transitions, and explanation style. It is not meant to infer personality, identity, ability, or private traits.

Inclusion condition

A preference enters the profile only after three similar interruption-and-repair observations, reducing one-click overinterpretation.

The Thing Itself: Multi-Agent Workflow

task -> generate -> stop -> interpret -> repair -> choose -> infer -> memory -> resume

System design

The system is a multi-agent loop built with AutoGen Core: generate, interrupt, interpret, repair, infer, remember. The prototype keeps generated text, repairs, explanation, and profile memory visible.

Synthetic Profile Data

150 simulated profiles	**50** steps each
5176 interruptions	**9–12** target items

Each fake user receives a common, rare personal, or mixed hidden writing-preference profile. The helper observes interruptions and repairs, then tries to recover repeated preferences.

Interpretation

Hidden preferences show the simulated distribution of writing needs across fake users.

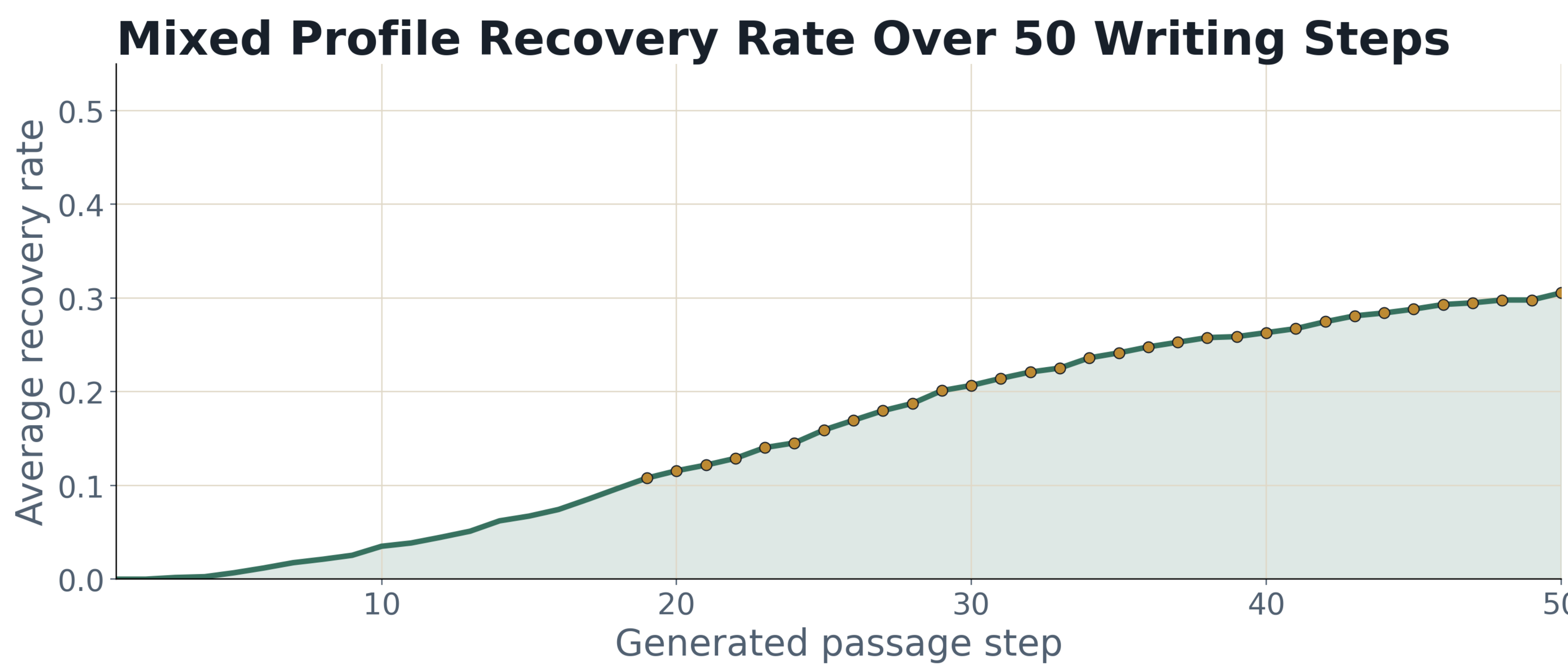
Recovered preferences show what the system actually learned after interruption and repair.

The overlap shows that repeated interruptions can recover meaningful writing-preference patterns.

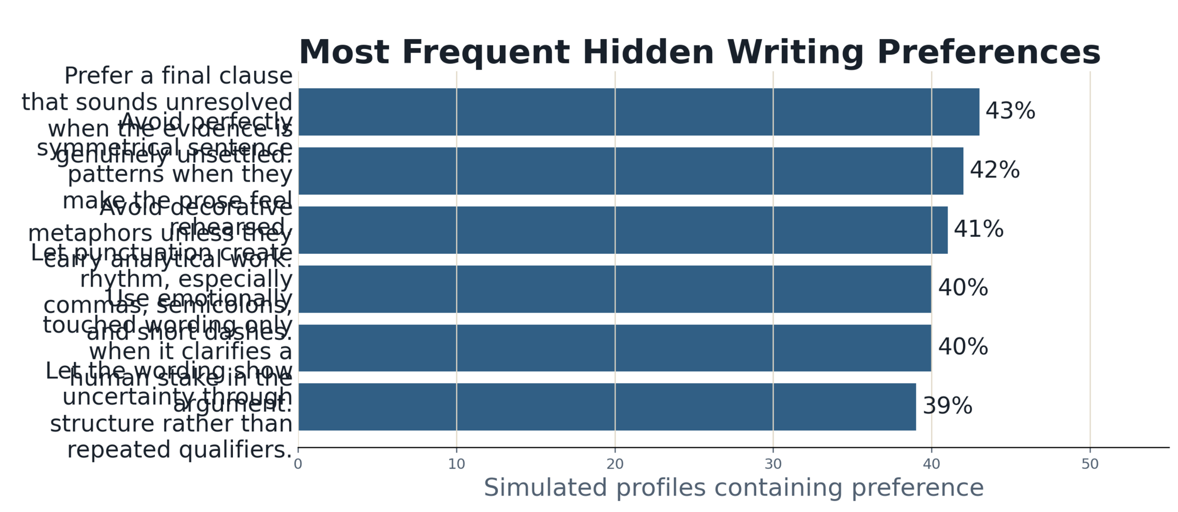
The result supports the design idea, but only for synthetic users and controlled interruption behavior.

Mixed Profile Recovery Rate

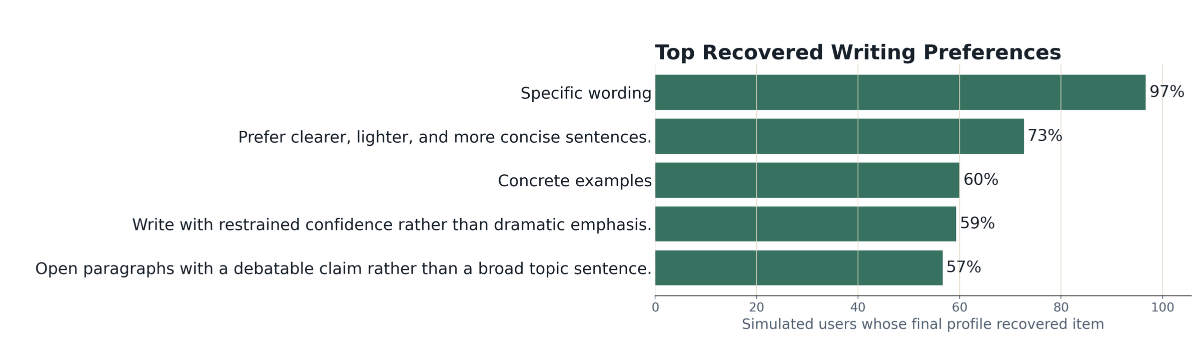
The full evaluation includes common, rare personal, and mixed hidden-profile groups. This poster shows the mixed-group result, where each profile combines rare style preferences with common academic-writing preferences.



Hidden Preferences



Top Recovered Preferences



Process Highlights

Iteration 1: I started with a generic writing helper, then made it interpret stopping points and generate replacements.
Iteration 2: I added an interpreter and rejected immediate profile updates, using a three-observation rule to reduce one-click overfitting.
Iteration 3: I generated random profiles and used an LLM to simulate the writing process, then summarized the data and recovery rate.

Reflection

Limit: the simulation shows the mechanism works, but it does not prove real writers interrupt consistently.
Next: test with real writers and add controls to approve, edit, or delete inferred preferences.

AI Disclosure

I designed the main system and used vibe coding for implementation support. AI assistance was used for coding, debugging, and code revision; generated edits were reviewed and approved.

References Leading To This Work

CoAuthor. Lee, M., Liang, P., and Yang, Q. (2022). Interaction traces reveal human-AI writing collaboration. `https://arxiv.org/abs/2201.06796`

GhostWriter. Yeh, C., Ramos, G., Ng, R., Huntington, K., and Banks, R. (2024). Personalization and agency in collaborative AI writing. `https://arxiv.org/abs/2402.08855`

GitHub: `https://github.com/Jerry241410/CMSC-FINAL.git` | **Runnable artifact:** `python writing.py` | **Data source:** `docs/poster_simulation.json`